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CLINICAL INVESTIGATION **Eye**

PROTON BEAM RADIOTHERAPY FOR UVEAL MELANOMA: RESULTS OF CURIE INSTITUT–ORSAY PROTON THERAPY CENTER (ICPO)

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Purpose: This study reports the results of proton beam radiotherapy based on a retrospective series of patients treated for uveal melanoma at the Orsay Center.
Methods and Materials: Between September 1991 and September 2001, 1,406 patients with uveal melanoma were treated by proton beam radiotherapy. A total dose of 60 cobalt Gray equivalent (CGE) was delivered in 4 fractions on 4 days. Survival rates were determined using Kaplan–Meier estimates. Prognostic factors were determined by multivariate analysis using the Cox model.
Results: The median follow-up was 73 months (range, 24–142 months). The 5-year overall survival and metastasis-free survival rates were 79% and 80.6%, respectively. The 5-year local control rate was 96%. The 5-year enucleation for complications rate was 7.7%. Independent prognostic factors for overall survival were age ($p < 0.0001$), gender ($p < 0.0003$), tumor site ($p < 0.0001$), tumor thickness ($p = 0.02$), tumor diameter ($p < 0.0001$), and retinal area receiving at least 30 CGE ($p = 0.003$). Independent prognostic factors for metastasis-free survival were age ($p = 0.0042$), retinal detachment ($p = 0.01$), tumor site ($p < 0.0001$), tumor volume ($p < 0.0001$), local recurrence ($p < 0.0001$), and retinal area receiving at least 30 CGE ($p = 0.002$). Independent prognostic factors for local control were tumor diameter ($p = 0.003$) and macular area receiving at least 30 CGE ($p = 0.01$). Independent prognostic factors for enucleation for complications were tumor thickness ($p < 0.0001$) and lens volume receiving at least 30 CGE ($p = 0.0002$).
Conclusion: This retrospective study confirms that proton beam radiotherapy ensures an excellent local control rate. Further clinical studies are required to decrease the incidence of postirradiation ocular complications. © 2006 Elsevier Inc.

Uveal melanoma, Proton beam radiotherapy, Local control, Survival, Enucleation, Ocular complication.

INTRODUCTION

Uveal melanoma is the most frequent ocular tumor but remains rare with a reported incidence of 6 to 7 cases per 100,000 inhabitants (1, 2). Recently, surgical enucleation was the standard treatment, but various eye-preserving treatment modalities have subsequently been developed, such as local resection (3–6), plaque brachytherapy (7, 8), external beam radiotherapy with photons using a linear accelerator (9), helium ions (10), and proton beam therapy (11–15). Proton beam radiotherapy is a conservative alternative to enucleation for the management of uveal melanoma, because a comparison between enucleation and radiotherapy does not reveal any significant difference in terms of overall survival and metastasis-free survival (16). In 1954, at the initiative of Irène Joliot-Curie, a synchrocyclotron particle accelerator was constructed by Philips at the Orsay University campus. It was initially devoted to nuclear physics research, but a phase of experimentation to develop medical applications was initiated in 1987 (17). Proton beam radiotherapy at Orsay, essentially for uveal melanoma, was performed for the first time in 1991 (18–20).

In this study, we will report the results of proton beam radiotherapy for uveal melanoma on a retrospective consecutive series of patients with a minimum follow-up of 2 years.

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